From Macro Regimes to Price Forecasts:
A Two-Stage Statistical Pipeline for Crude-Oil Prediction

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Abstract

This report develops a two-stage statistical pipeline for short-horizon crude-oil price forecasting. **Stage 1** identifies macroeconomic regimes endogenously by applying K-means clustering to five standardised macro-financial features (VIX, 10Y–2Y yield spread, Economic Policy Uncertainty index, Geopolitical Risk Daily index, gold-to-oil ratio) over the period 2000–2025. Four interpretable, highly persistent regimes emerge – calm expansion, late-cycle / moderate stress, acute crisis, and policy-uncertainty dominated – whose historical mapping aligns closely with well-known episodes (2008 GFC, 2011 Eurozone crisis, 2020 COVID crash, 2020–2022 post-pandemic policy uncertainty). **Stage 2** conditions on the large late-cycle regime (the "red" regime, 2,443 trading days) and fits **HYDRA**, a nine-stage hybrid pipeline combining PCA, K-means, Fisher LDA, Naive Bayes, three regime-specialised regressors (SVR, GBM, MLP), four global regressors, and a gradient-boosted meta-stacker. On a held-out December 2024 test set, HYDRA achieves RMSE = \$0.813/bbl and MAE = \$0.627/bbl, essentially tying the best single global model (Random Forest, RMSE = \$0.775). We conclude that regime conditioning yields interpretable and historically meaningful partitions of the sample, while stacking on top of a strong tree ensemble improves *robustness* more than it improves headline accuracy on short test windows with highly autoregressive targets.

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1 Introduction

Crude-oil prices are driven by a complex mixture of supply-side fundamentals, demand-side macroeconomic conditions, and risk-driven financial flows. A single statistical model that treats all trading days identically is therefore mis-specified: the elasticities that govern prices in calm periods differ from those that govern prices during crises or periods of elevated policy uncertainty.

The project addresses this heterogeneity through a two-stage conditioning strategy:

Stage 1. *Macro-regime identification.* Instead of defining regimes *ex ante* (e.g. NBER recessions, or VIX > 30), we apply K-means to a panel of five macro-financial features and interpret the resulting clusters *post hoc*. This yields four persistent regimes covering the whole 2000–2025 sample.

Stage 2. *Conditional forecasting.* Trading days belonging to the largest non-expansion regime – dubbed the "red" / late-cycle regime – are extracted and merged with a commodity-price panel from Investing.com. A nine-stage ensemble pipeline (HYDRA) is then trained to predict daily WTI crude oil on this sub-sample, with December 2024 held out for evaluation.

Section 2 describes the two datasets used. Part I (I–5.6) documents the regime classification. Part II (II–11) documents the HYDRA pipeline. Section 12 summarises the overall findings.

2 Data

This study draws on two independently constructed datasets. The first is a multi-source panel of financial market prices and macro-uncertainty indices used for regime classification. The second combines daily equity and commodity prices with the Geopolitical Risk (GPR) index to form the input to the crude-oil forecasting stage. Both datasets span the post-2000 period, covering the dot-com collapse, the Global Financial Crisis, the European sovereign-debt crisis, the COVID-19 pandemic, and the post-2022 inflation / geopolitical-tension episodes.

2.1 Dataset 1: Multi-Source Regime Classification Panel

The first dataset is a daily panel spanning **1 January 2000 to 31 December 2025**, covering all U.S. trading days (Monday–Friday). It integrates nine macro-financial variables obtained from four independent sources.

Data sources. Asset price series are retrieved via the `quantmod` interface to Yahoo Finance; U.S. Treasury yields are obtained from the Federal Reserve Economic Data (FRED) API; the Economic Policy Uncertainty (EPU) index is sourced from Baker, Bloom and Davis (policyuncertainty.com); and the daily Geopolitical Risk index (GPRD) is sourced from Caldara and Iacoviello (matteoiacoviello.com). A detailed list of variables, tickers and sources is reported in Table 1.

Table 1: Variables in the Regime Classification Panel.

Variable	Ticker / ID	Source	Description
CBOE Volatility Index	`^VIX`	Yahoo Finance	Implied 30-day S&P 500 volatility.
WTI Crude Oil (Front)	`CL=F`	Yahoo Finance	Front-month WTI futures price (USD/bbl).
S&P 500 Index	`^GSPC`	Yahoo Finance	Large-cap U.S. equity benchmark.
Gold Futures	`GC=F`	Yahoo Finance	Front-month COMEX gold (USD/oz).
U.S. Dollar Index	`DX-Y.NYB`	Yahoo Finance	Trade-weighted USD vs. major currencies.
10-Year Treasury Yield	`DGS10`	FRED	Constant-maturity 10-year yield (%).
2-Year Treasury Yield	`DGS2`	FRED	Constant-maturity 2-year yield (%).
Economic Policy Uncertainty	EPU	Baker et al.	News-based monthly U.S. policy-uncertainty index.
Geopolitical Risk (Daily)	GPRD	Caldara & Iacoviello	News-based daily geopolitical-risk index.

Alignment and missing-value treatment. All series are aligned on a common trading-day calendar. The EPU index, which is published at monthly frequency, is linearly interpolated to daily observations using `na.approx`. After merging, missing values caused by holidays and asynchronous release calendars are handled in two passes: (i) last-observation-carried- forward to propagate the most recent known value across gaps, and (ii) a backward fill to address any leading missing values at the start of a series. This procedure preserves the release-time information structure while producing a rectangular panel suitable for time-series modelling.

Derived variables and feature engineering. A 10Y–2Y term-spread series is constructed from the two Treasury yields and serves as a conventional recession indicator. From the aligned panel, we additionally compute: log returns (S&P 500, crude oil, gold); 30-day rolling standard deviations of those returns and a 30-day rolling mean of the VIX; a gold-to-oil ratio (safe-haven vs. cyclical demand); a VIX-to-EPU ratio (market-implied vs. policy- driven uncertainty); and two binary regime indicators (yield-curve-inversion flag, and VIX > 30 stress flag). The final dataset contains 6,783 trading-day observations across 19 columns and is exported as `regime_data_2000_2025.csv`.

2.2 Dataset 2: Equity/Commodity Prices and the Geopolitical Risk Index

The second dataset combines daily commodity and equity prices with the Geopolitical Risk (GPR) index of Caldara and Iacoviello and is used as the input to the crude-oil forecasting stage (7).

Data sources. Daily closing prices for crude oil, gas oil, heating oil, and natural gas are obtained from **Investing.com**, which provides standardised end-of-day quotes adjusted for trading-calendar conventions. The GPR and VIX series are also included. The file is `combined_prices.csv` and contains 5,920 trading-day observations across seven columns.

Merging. Dataset 2 is inner-joined with Dataset 1 on the common date field, retaining only dates on which both data sources are observed. The resulting merged panel (`merged_market_data.csv`,

24 columns) is the master data used downstream. From this master panel we extract the subset of trading days whose cluster assignment from Part I equals the red regime; this yields `merged_red_regime_data.csv`, which is the input to the HYDRA pipeline.

Reproducibility. All data collection and cleaning is implemented in R (version 4.x) using `quantmod`, `fredr`, `readxl`, `dplyr`, `zoo` and `roll`.

Part I
Macro-Regime Classification

3 Objective and Methodology (Part I)

3.1 Objective

Financial markets exhibit regime-dependent behaviour: the joint distribution of returns, volatilities, and macro-uncertainty measures differs materially between expansions, crises, and periods of elevated policy or geopolitical stress. Rather than defining regimes *ex ante* using ad-hoc thresholds (e.g. NBER recession dates, or VIX > 30), we identify them *endogenously* using K-means on a panel of macro-financial indicators covering 2000–2025. The resulting regime labels are then used in Part II to construct a conditional sub-sample of late-cycle-like observations for downstream commodity- price modelling.

3.2 Feature set

The clustering uses five standardised features chosen to capture complementary dimensions of market stress and macro uncertainty:

- **VIX** – implied equity-market volatility;
- **10Y–2Y yield spread** – a classical business-cycle / recession indicator;
- **EPU index** – news-based measure of domestic policy risk;
- **GPRD index** – Caldara–Iacoviello daily geopolitical-risk measure;
- **Gold-to-oil ratio** – a cross-asset proxy that rises when safe-haven demand dominates cyclical energy demand.

Each feature is standardised to zero mean and unit variance prior to clustering so that the Euclidean distance used by K-means is not dominated by any single variable's scale.

3.3 Choice of K: elbow criterion

The optimal number of clusters is selected via the elbow method, which plots the total within-cluster sum-of-squares (WSS) against K and identifies the value beyond which the marginal gain from adding another cluster becomes small. The curve is displayed in Figure 1.

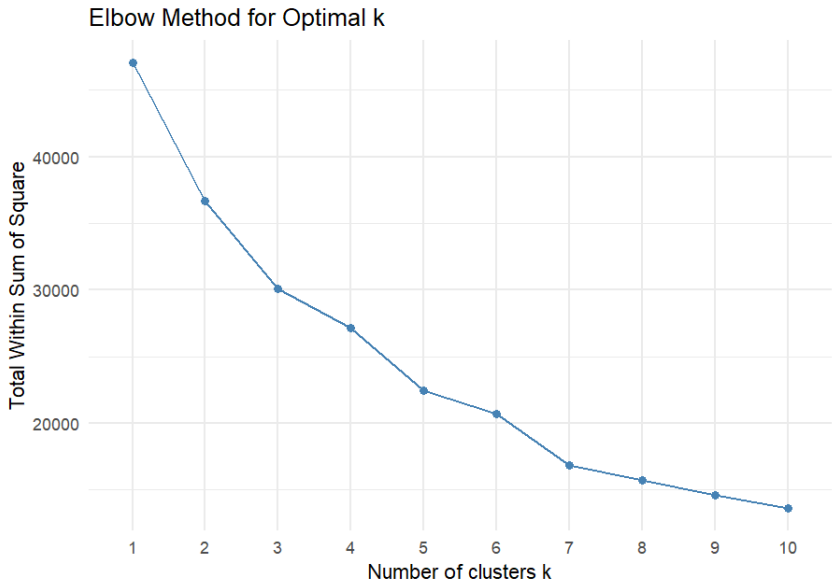

Figure 1: Total within-cluster sum of squares as a function of the number of clusters K. The curve shows a clear kink around $K = 4$, after which additional clusters yield only modest reductions in WSS.

The steepest reductions in WSS occur from $K = 1$ to $K = 4$, with the curve flattening noticeably thereafter. We therefore fix $K = 4$, balancing parsimony against interpretability.

3.4 K-means estimation

K-means is run with `nstart = 25` random initialisations to mitigate sensitivity to starting values. The algorithm assigns each trading day $t \in \{2000\text{-}01\text{-}03, \ldots, 2025\text{-}12\text{-}31\}$ to exactly one of four regime clusters, denoted $\mathcal{R}_1, \ldots, \mathcal{R}_4$.

4 Regime Classification Results

4.1 Cluster profiles

The centroid values of the five input features, together with the number of trading days assigned to each cluster, are reported in Table 2.

Table 2: Regime centroids and frequencies.

Regime	Colour	VIX	Yield Spread	EPU	GPRD	Gold/Oil	Days
$\mathcal{R}_1$	Red	16.6	0.54	164.1	115.9	24.5	2,443
$\mathcal{R}_2$	Blue	36.9	1.84	170.0	174.9	15.5	638
$\mathcal{R}_3$	Green	18.2	1.39	103.0	106.4	12.1	3,204
$\mathcal{R}_4$	Purple	23.9	0.51	369.0	127.0	52.0	437

The four centroids differ markedly across every feature dimension, which is a first indication that the clusters capture economically distinct states rather than arbitrary partitions of a single underlying distribution.

4.2 Low-dimensional visualisation

To verify cluster separation, the standardised feature matrix is projected onto its first two principal components, which together explain $34.1\% + 31.4\% = 65.5\%$ of the total variance. The scatter is shown in Figure 2.

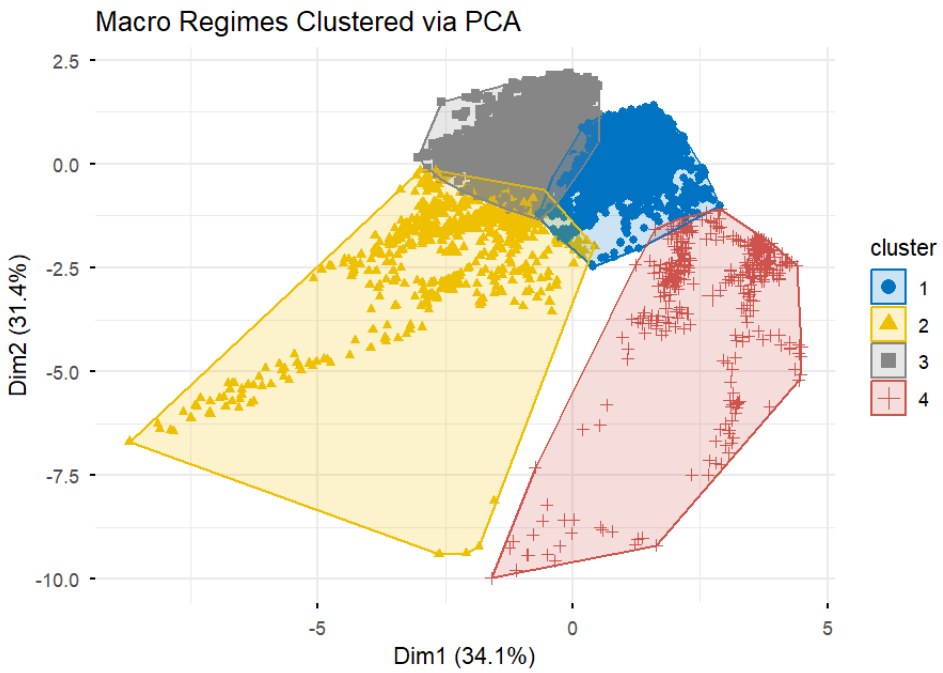

Figure 2: Regime clusters in the space of the first two principal components. The four convex hulls overlap only marginally, confirming that the clusters occupy distinct regions of feature space.

Cluster 3 (grey in the PCA plot / green in the temporal plots) occupies the upper-centre region and corresponds to low-stress observations; cluster 2 (yellow / blue) sits in the lower-left, reflecting an extreme combination of high VIX and high GPRD; cluster 4 (red symbols in the PCA plot / purple in the temporal plots) extends along the right-hand edge, driven by extreme EPU values; and cluster 1 (blue in the PCA plot / red in the temporal plots) occupies a central transitional zone between calm and stress regions.

4.3 Temporal distribution

Figure 3 overlays the regime labels on the S&P 500 price series, while Figure 4 presents the same information as a compact "barcode" strip that emphasises regime persistence and transitions.

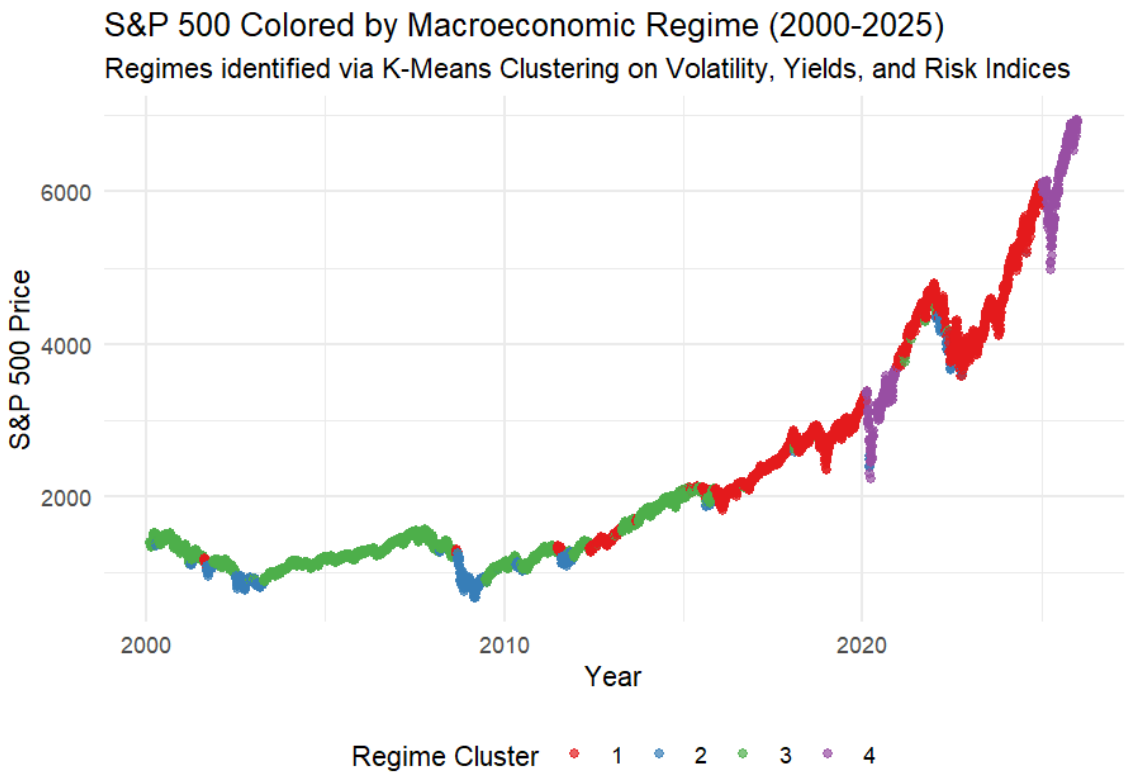

Figure 3: S&P 500 daily close coloured by cluster membership (2000–2025).

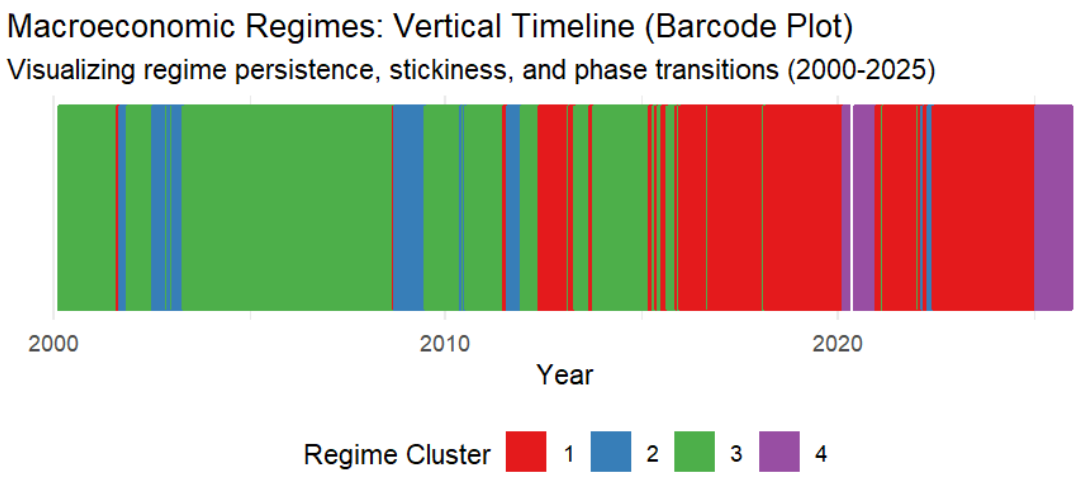

Figure 4: Vertical-bar ("barcode") representation of the regime time series. Each column represents one trading day.

Two structural features are immediately visible. First, regimes are *highly persistent*: the plot exhibits long contiguous blocks of a single colour rather than rapid flicker, which is consistent with the slow-moving nature of the underlying macro fundamentals. Second, there is a clear *regime shift around 2015*: the pre-2015 sample is dominated by green ($\mathcal{R}_3$), whereas the post-2015 sample is dominated by red ($\mathcal{R}_1$) with purple ($\mathcal{R}_4$) episodes concentrated in 2020–2022 and 2025.

5 Regime Interpretation

5.1 $\mathcal{R}_3$ (Green) — Calm Expansion

This is the modal regime, covering roughly $\frac{3204}{6722} \approx 47.7\%$ of trading days. It is characterised by low volatility (VIX ≈ 18), a *steep* yield curve (10Y–2Y spread ≈ 1.4 pp), low policy uncertainty (EPU ≈ 103), low geopolitical risk (GPRD ≈ 106) and a compressed gold-to-oil ratio. Historically, it maps onto the 2003–2007 pre-crisis bull market, the 2013–2015 low-volatility "goldilocks" period, and most of the 2010–2012 mid-cycle phase. It is, in effect, the textbook expansion regime.

5.2 $\mathcal{R}_2$ (Blue) — Acute Crisis

Despite containing only $\frac{638}{6722} \approx 9.5\%$ of observations, this cluster captures the most economically consequential episodes in the sample. Its centroid combines an extremely elevated VIX (36.9), the steepest yield spread of all four regimes (1.84 pp, consistent with aggressive Fed easing in response to stress), and the highest geopolitical risk reading (GPRD = 175). On the historical timeline, the blue stripes align tightly with:

- the 2001–2002 dot-com collapse and 9/11 aftermath,
- the 2008–2009 Global Financial Crisis,
- the 2011–2012 Eurozone sovereign-debt crisis and U.S. debt-ceiling episode,
- the March–April 2020 COVID-19 market crash.

We therefore label $\mathcal{R}_2$ the *acute crisis* regime.

5.3 $\mathcal{R}_1$ (Red) — Late-Cycle / Moderate Stress

This is the second-largest cluster ($\approx 36.3\%$ of days) and is defined by moderate VIX (16.6), a nearly flat yield curve (spread = 0.54 pp), elevated-but-not-extreme policy uncertainty (EPU = 164), and an elevated gold-to-oil ratio (24.5). The flat yield curve combined with only moderately elevated uncertainty is the canonical signature of a *late-cycle* environment: growth is still positive but the term structure signals that the cycle is maturing. Temporally, $\mathcal{R}_1$ dominates the post-2015 period and reappears in 2023–2024 after the 2022 inflation shock, consistent with this interpretation. **Part II of this report is built entirely on the $\mathcal{R}_1$ sub-sample.**

5.4 $\mathcal{R}_4$ (Purple) — Policy-Uncertainty Dominated

The smallest cluster ($\approx 6.5\%$ of days) is distinguished almost entirely by one feature: the EPU index centroid of 369 is more than *double* that of any other regime. VIX is elevated (23.9) but not at crisis levels, the yield curve is flat, and the gold-to-oil ratio is extremely high (52.0), indicating that commodity markets are pricing in weak cyclical demand simultaneously with strong safe-haven flows. The purple stripes in Figure 4 concentrate in 2020–2022 – covering

the U.S. election, the post-pandemic fiscal-policy debates, and the inflation-driven monetary-policy repricing – and reappear in 2025. We therefore label this the *policy-uncertainty* regime. Importantly, it is *distinct from* the acute-crisis regime ($\mathcal{R}_2$): markets in $\mathcal{R}_4$ are stressed but not panicked, and the stress is sourced from the policy channel rather than from geopolitical or financial-market dislocation.

5.5 Summary of Part I

The four clusters admit a coherent economic interpretation that is *not imposed* on the data: calm expansion ($\mathcal{R}_3$), late-cycle / moderate stress ($\mathcal{R}_1$), acute crisis ($\mathcal{R}_2$), and policy-uncertainty dominated ($\mathcal{R}_4$). The fact that the two stress regimes ($\mathcal{R}_2$ and $\mathcal{R}_4$) separate cleanly along the VIX/GPRD axis versus the EPU axis is itself an interesting empirical finding: financial-market stress and policy uncertainty are correlated but not redundant, and the data-driven clustering recovers that distinction without guidance.

5.6 Sub-sample extraction for Part II

For the second stage of the project we extract the set of dates assigned to regime $\mathcal{R}_1$ (red) and inner-join them with the commodity-price panel `combined_prices.csv`. Restricting commodity models to this sub-sample provides a training set that is internally homogeneous in its macro-uncertainty conditions and should therefore reduce the parameter instability associated with regime mixing. The resulting file, `merged_red_regime_data.csv`, covers 31 October 2000 to 27 December 2024 and contains 2,443 trading days.

Part II
The HYDRA Forecasting Pipeline

6 Overview and Architecture

Part II documents **HYDRA**, a nine-stage hybrid pipeline that predicts the next-day close of WTI crude oil on the red-regime sub-sample constructed in 5.6. The intuition is two-fold. First, even within the late-cycle macro regime, trading days differ substantially in their level of market stress; a finer three-way partition is therefore learned from the stress indicators (VIX, EPU, GPRD). Second, different statistical learners excel in different conditions, so each micro-regime is assigned its own *expert* regressor, and those experts are then blended with four *global* regressors through a gradient-boosted meta-stacker. The architecture is summarised in Figure 5.

HYDRA pipeline (nine stages).

1. Data loading, red-regime filter, lag construction.
2. *PCA* on three standardised stress features.
3. *K-means* ($K = 3$) on the stress features; soft memberships via softmax.
4. *Fisher LDA* using the K-means labels as supervision.
5. *Naive Bayes* on the LDA-scaled features for regime posteriors.
6. Three *regime-expert regressors*: SVR (High-Stress), GBM (Med-Stress), MLP (Low-Stress). Predictions combined by soft-membership weights.
7. Four *global regressors*: SVR, MLP, Random Forest, GBM, evaluated by 5-fold rolling-origin time-series cross-validation.
8. *Meta-stacker*: gradient-boosted regression on the five Level-1 predictions.
9. Evaluation on the held-out December 2024 test set.

Figure 5: Architecture of the HYDRA pipeline.

7 Data and Exploratory Analysis (Part II)

7.1 Scope and target

The conditional sample contains 2,443 trading days assigned to the red macro regime between 31 October 2000 and 27 December 2024 (the sub-sample extracted in 5.6). The prediction target is the daily WTI crude-oil close. The final month of the sample (December 2024, 19 trading days) is held out as the test set; the remainder forms the training set.

Feature groups used downstream are:

- **Stress features:** `vix_price`, `gprd_index`, `epu_index`.
- **Regression features:** the three stress features, plus `sp500_price` and four autoregressive lags of crude price at horizons $\{1, 2, 3, 5\}$ trading days.

7.2 Univariate distributions

Table 3 reports the univariate summary statistics of the key variables on the training set.

Table 3: Training-set summary statistics (red-regime sample).

Variable	Mean	SD	Min	Median	Max
Crude oil (USD)	65.78	16.29	26.21	65.97	109.71
VIX	16.08	4.14	9.14	15.36	37.21
GPRD index	114.19	45.30	9.49	106.86	413.46
EPU index	159.89	38.46	76.84	155.94	310.14
S&P 500	3,165.96	1,227.64	1,085.78	2,821.16	6,032.38
Gold (USD/oz)	1,588.36	389.12	271.60	1,507.40	2,788.50

The corresponding histograms are shown in Figure 6. Three features are clearly multi-modal on the sample: crude oil (a cluster near $50 corresponding to the 2015–2017 shale glut and another near $75 corresponding to the 2018–2023 tight balance), the S&P 500 (reflecting the pre- and post- 2020 level shifts), and gold (a left peak around $1,200 and a right peak above $1,800). Stress indicators (VIX, GPRD, EPU) are all right-skewed, consistent with the conditioning on a non-crisis regime: extreme tails are truncated away.

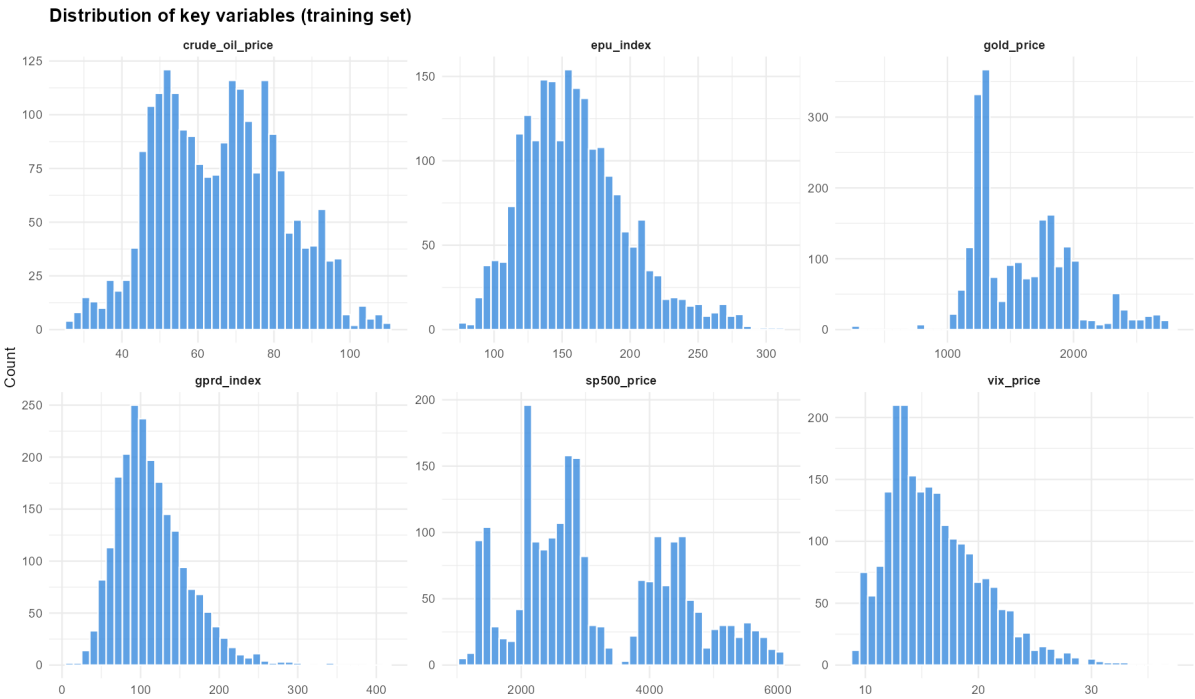

Figure 6: Histograms of the six key input variables over the training set. Crude oil, S&P 500, and gold are visibly multi- modal; the stress indicators are unimodal and right-skewed.

The long-run trajectory of crude oil on the red-regime sample is plotted in Figure 7. The red dashed line marks the December 2024 train/test boundary.

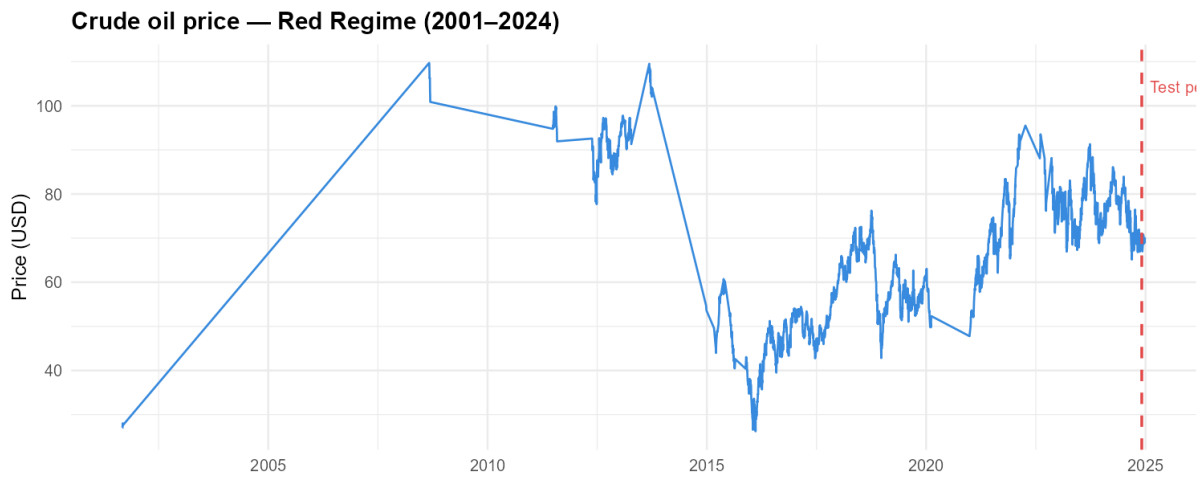

Figure 7: Crude-oil price over the red-regime sample (2001–2024). The straight segments reflect date gaps where consecutive red- regime days are not adjacent in calendar time.

7.3 Cross-sectional correlations

Figure 8 presents the Pearson correlation matrix of the key features. Three observations deserve emphasis:

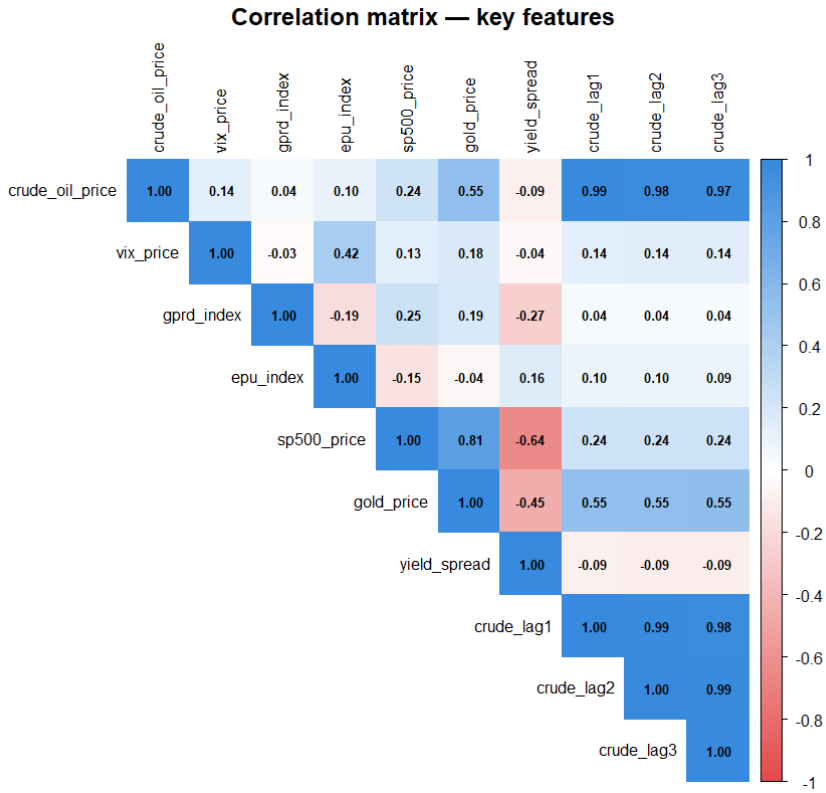

Figure 8: Pearson correlations among key features (training set).

- *Autoregressive persistence dominates.* The correlation between crude oil and its own first lag is 0.99, and remains above 0.97 at a five-day horizon. Any credible baseline must therefore exploit lagged prices; any "improvement" that does not outperform a lag-1 predictor is suspect.

- *Contemporaneous financial-market correlations with crude are mild.* The strongest such link is with gold ($r = 0.55$), followed by the S&P 500 ($r = 0.24$). VIX, GPRD, EPU and the yield spread are all individually weak predictors of crude ($|r| \leq 0.14$), motivating the use of nonlinear regressors rather than a linear model.

- *Macro block signs are canonical.* The S&P 500 loads positively on gold (0.81) and negatively on the yield spread (-0.64); gold loads negatively on the yield spread (-0.45). This cross-validates that our red regime cleanly isolates a late-cycle environment.

7.4 Stress indicators and conditional scatter

Figure 9 plots VIX, GPRD and EPU over the sample; Figure 10 then shows scatter plots of crude against each of the four main predictors, overlaid with a LOESS smoother. Although we conditioned on a non-crisis macro regime, there is still substantial within-regime variation in the stress indicators: VIX ranges from roughly 9 to 37, GPRD from 9 to 413, and EPU from 77 to 310. This within-regime variation is what the downstream micro-regime clustering exploits. The scatter plots make clear that the relationships are *nonlinear and non-monotonic*: crude rises modestly with VIX up to ≈ 18 and then falls; it is essentially flat in EPU except at the tails; and the crude–S&P relationship is sharply U-shaped because low-S&P regions include both the 2008 spike and the 2020 crash. A linear regression specification would be inadequate.

Figure 9: VIX, GPRD and EPU over the red-regime sample.

Figure 10: Crude oil vs. each of the four main predictors. Red curves are LOESS smoothers.

8 Theoretical Background

We briefly review the statistical components used in HYDRA. The unifying design principle is that each unsupervised step (8.1–8.4) produces engineered features that are appended to the raw regression matrix, while each supervised step (8.5–8.7) produces a Level-1 prediction that the meta-stacker (8.8) consumes.

8.1 Principal Component Analysis

Given a centred and scaled data matrix $\boldsymbol{X} \in \mathbb{R}^{n \times p}$, PCA computes the eigendecomposition of the sample covariance $\boldsymbol{S} = \frac{1}{n-1}\boldsymbol{X}^\top \boldsymbol{X}$ and projects $\boldsymbol{X}$ onto the leading eigenvectors:

$$\boldsymbol{Z} = \boldsymbol{X}\boldsymbol{V}_k, \qquad \boldsymbol{S}\boldsymbol{V} = \boldsymbol{V}\boldsymbol{\Lambda}, \qquad \boldsymbol{V}_k \in \mathbb{R}^{p \times k}.$$

The columns of $\boldsymbol{Z}$ are the principal components; the fraction of variance they explain is $\lambda_j / \sum_{i=1}^{p} \lambda_i$. We apply PCA to the three standardised stress features and retain the first two components.

8.2 K-means clustering

K-means partitions observations $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n \in \mathbb{R}^p$ into K disjoint clusters $C_1, \ldots, C_K$ that minimise the within-cluster sum of squares:

$$\min_{\{C_k\},\{\boldsymbol{\mu}_k\}} \sum_{k=1}^{K} \sum_{\boldsymbol{x}_i \in C_k} \|\boldsymbol{x}_i - \boldsymbol{\mu}_k\|_2^2.$$

The algorithm alternates an assignment step, $C_k = \{i : \|\boldsymbol{x}_i - \boldsymbol{\mu}_k\|_2 \le \|\boldsymbol{x}_i - \boldsymbol{\mu}_\ell\|_2 \ \forall \ell\}$, with an update step, $\boldsymbol{\mu}_k = |C_k|^{-1} \sum_{i \in C_k} \boldsymbol{x}_i$. In Part II we use $K = 3$ and produce *soft* memberships via a softmax over negative distances,

$$w_{ik} \;=\; \frac{\exp(-\|\boldsymbol{x}_i - \boldsymbol{\mu}_k\|_2)}{\sum_{\ell=1}^{K} \exp(-\|\boldsymbol{x}_i - \boldsymbol{\mu}_\ell\|_2)}, \tag{1}$$

which serve as the blending weights for the regime-expert regressors.

8.3 Fisher's Linear Discriminant Analysis

Given class labels $g \in \{1, \ldots, K\}$, LDA finds a linear projection $\boldsymbol{w} \in \mathbb{R}^p$ that maximises the between-class to within-class variance ratio

$$J(\boldsymbol{w}) = \frac{\boldsymbol{w}^\top \boldsymbol{S}_B \boldsymbol{w}}{\boldsymbol{w}^\top \boldsymbol{S}_W \boldsymbol{w}},$$

with $\boldsymbol{S}_B$ and $\boldsymbol{S}_W$ the between- and within-class scatter matrices. Solving the generalised eigenproblem $\boldsymbol{S}_B \boldsymbol{w} = \lambda \boldsymbol{S}_W \boldsymbol{w}$ yields up to $K-1$ discriminant directions; we retain the first two. Using the K-means labels as the supervision target re-projects the feature space to maximise separability of the very clusters K-means just discovered, typically giving cleaner decision boundaries than PCA's variance-maximising projection.

8.4 Naive Bayes

Naive Bayes estimates the posterior class probability

$$\Pr(G = k \mid \boldsymbol{x}) \;\propto\; \pi_k \prod_{j=1}^{p} f_j(x_j \mid G = k), \tag{2}$$

under the conditional-independence assumption that features are independent within class. With Gaussian class-conditional densities, $f_j(\cdot \mid k) = \mathcal{N}(\mu_{kj}, \sigma_{kj}^2)$, the parameters reduce to per-class means and variances. Even when the conditional-independence assumption is violated, the resulting class posteriors remain useful as *features*, because they compress multivariate information into K calibrated probabilities.

8.5 Support Vector Regression

ε-SVR finds a function $f(\boldsymbol{x}) = \boldsymbol{w}^\top \phi(\boldsymbol{x}) + b$ that minimises

$$\tfrac{1}{2}\|\boldsymbol{w}\|_2^2 + C \sum_{i=1}^{n} L_\varepsilon(y_i, f(\boldsymbol{x}_i)), \qquad L_\varepsilon(y, \hat{y}) = \max\{0,\, |y - \hat{y}| - \varepsilon\},$$

where L_ε is the ε-insensitive loss and ϕ is an implicit feature map induced by a positive-definite kernel $k(\boldsymbol{x}, \boldsymbol{x}') = \langle \phi(\boldsymbol{x}), \phi(\boldsymbol{x}') \rangle$. We use the radial-basis kernel, $k(\boldsymbol{x}, \boldsymbol{x}') = \exp(-\gamma \|\boldsymbol{x} - \boldsymbol{x}'\|_2^2)$. SVR is assigned to the high-stress micro-regime because its margin-based formulation tends to be robust to outliers, which dominate the high-VIX sample.

8.6 Random Forests and Gradient Boosting

A Random Forest *Breiman2001 is a bagged ensemble of decorrelated CART regression trees:

$$\hat{f}_{\mathrm{RF}}(\boldsymbol{x}) = \frac{1}{B} \sum_{b=1}^{B} T_b(\boldsymbol{x}),$$

where each T_b is grown on a bootstrap sample with a random subset of features considered at each split. Gradient Boosting, by contrast, is a forward stagewise additive model,

$$\hat{f}^{(m)}(\boldsymbol{x}) = \hat{f}^{(m-1)}(\boldsymbol{x}) + \eta\, h_m(\boldsymbol{x}),$$

where each new tree h_m is fit to the negative gradient of the loss at the current prediction. We use $\eta = 0.05$–0.08 and 150–200 trees. The two ensembles are complementary: RF reduces variance, GBM reduces bias.

8.7 Multilayer Perceptron

A single-hidden-layer MLP with H units implements

$$f(\boldsymbol{x}) = \boldsymbol{w}_2^\top \sigma(\boldsymbol{W}_1 \boldsymbol{x} + \boldsymbol{b}_1) + b_2,$$

with σ the logistic activation. We fix $H = 8$ with weight-decay regularisation $\lambda = 10^{-3}$ and optimise by BFGS (the `nnet` default) for up to 500 iterations. Targets are min–max scaled to $[0, 1]$ so that the output layer remains in the range of the activation.

8.8 Time-series cross-validation and stacking

Because trading days are ordered and autocorrelated, we evaluate global regressors with rolling-origin time-series cross-validation: for $k = 1, \ldots, 5$, the k-th training fold uses observations 1 through $k \cdot \lfloor n/6 \rfloor$ and the validation fold is the next block of the same size. Mean cross-validated RMSE is reported.

Stacking *Wolpert1992 combines M Level-1 predictions $\hat{y}^{(1)}, \ldots, \hat{y}^{(M)}$ through a Level-2 meta-learner trained to minimise $\sum_i (y_i - g(\hat{y}_i^{(1)}, \ldots, \hat{y}_i^{(M)}))^2$. We take g to be a gradient-boosted regression tree ensemble with 100 trees and depth 3, which allows non-linear and interaction-based combinations of the Level-1 predictions without requiring us to specify weights manually.

9 HYDRA Pipeline Implementation

9.1 Stage 1 – Preprocessing

After filtering to red-regime dates and dropping two duplicate columns inherited from the merge (Dataset 1's `VIX_price` and `GPRD`), four autoregressive lags of crude are generated at horizons $\{1, 2, 3, 5\}$ and rows with any missing values are removed. The training set ends on 29 November 2024; the test set is the 19 trading days of December 2024.

9.2 Stage 2 – PCA on stress features

PCA is fitted on the three standardised stress features from the training set. The first two components explain $49.1\% + 32.7\% = 81.8\%$ of the variance (Figure 11).

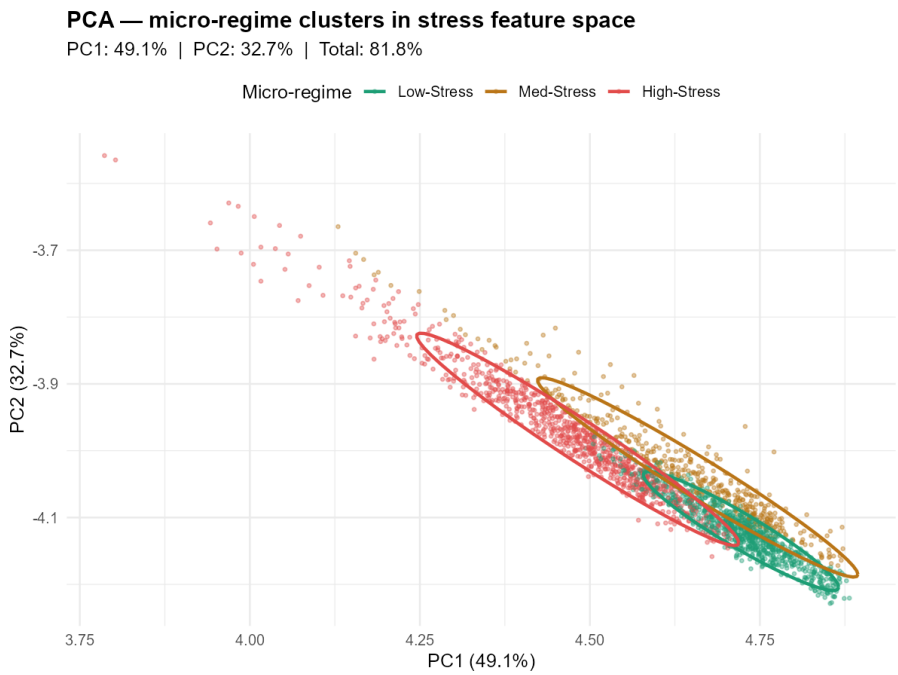

Figure 11: PCA projection of the training set onto the first two principal components of the three stress features. Colour denotes the micro-regime assigned by K-means.

The three clusters form a clear gradient along PC1: the Low-Stress cluster concentrates at the upper-right, the High-Stress cluster at the lower-left, and Med-Stress occupies the transition band. This validates $K = 3$ as a reasonable choice: the clusters align with a single dominant direction of variation in stress space. They do, however, overlap non-trivially in the 2D projection, which is why we use soft memberships rather than hard labels in the expert blend.

9.3 Stage 3 – K-means micro-regimes

K-means with $K = 3$ and `nstart`= 20 partitions the training sample by VIX-level into Low-/Med-/High-Stress. Cluster labels are assigned post-hoc in order of mean VIX so that the labelling is stable across runs. Soft memberships w_{ik} are computed via the softmax (1) and are the blending weights for the expert ensemble.

9.4 Stage 4 – Fisher LDA

LDA is fitted on the eight standardised regression features using the K-means labels as supervision. The top two LDA directions are retained and appended as engineered features. This addresses a subtle but important issue: K-means only sees the stress features, so its cluster labels carry information about VIX/EPU/GPRD only. LDA re-expresses *all eight* regression features in terms of the two directions that best separate those clusters, so the downstream models receive a second view of the data that is explicitly organised around micro-regime structure.

9.5 Stage 5 – Naive Bayes posteriors

Naive Bayes with Gaussian class-conditionals is fitted on the LDA-scaled features against the micro-regime labels. The three posterior probabilities are appended as features. Bayes training accuracy on the regime labels is reported by the pipeline and is used only as a sanity check, not as a downstream prediction.

9.6 Stage 6 – Regime-expert regressors

The eight raw regression features, together with the two PCs, the two LD coordinates and the three Naive Bayes posteriors, form a $p = 15$ augmented feature matrix $\boldsymbol{X}$. Three expert regressors are fitted:

Regime	Regressor	Hyperparameters
High-Stress	SVR (RBF)	cost = 200, $\varepsilon = 0.3$
Med-Stress	GBM	150 trees, depth 4, $\eta = 0.08$
Low-Stress	MLP	8 hidden units, decay 10^{-3}, 500 iter

For any new observation $\boldsymbol{x}_\star$, the expert blend forms

$$\hat{y}_\star^{\text{expert}} = \sum_{k=1}^{3} w_{\star k}\, \hat{f}_k(\boldsymbol{x}_\star), \tag{3}$$

where $w_{\star k}$ is the soft membership of $\boldsymbol{x}_\star$ in regime k and $\hat{f}_k$ is the corresponding expert.

9.7 Stage 7 – Global regressors

Four global regressors (SVR, MLP, RF, GBM) are fitted on the full training set with 5-fold rolling-origin CV reported on the training set. Hyperparameters follow the pipeline defaults in `02_HYDRA_Pipeline.R`.

9.8 Stage 8 – Meta-stacker

The five Level-1 predictions (expert blend, SVR, MLP, RF, GBM) are stacked into a meta-feature matrix and a final gradient-boosted model is fitted:

$$\hat{y}_\star^{\text{HYDRA}} = g_{\text{GBM}}\big(\hat{y}_\star^{\text{expert}},\ \hat{y}_\star^{\text{SVR}},\ \hat{y}_\star^{\text{MLP}},\ \hat{y}_\star^{\text{RF}},\ \hat{y}_\star^{\text{GBM}}\big)\,,$$

with 100 trees at depth 3 and $\eta = 0.05$. A known pitfall of this design is that the Level-1 predictions used to train the meta-stacker are *in-sample* predictions from models that have seen the training data, which will overstate their correlation with the target and bias the meta-weights. Proper out-of-fold stacking would mitigate this; we return to this point in 11.

10 HYDRA Results and Interpretation

10.1 Micro-regime structure

Figure 12 shows the micro-regime assignment of each training day over time, coloured by cluster. High-Stress days concentrate in the 2008, 2011–2012, and 2014 periods and reappear during the 2022 oil shock. Med-Stress and Low-Stress days dominate the post-2015 sample. The clustering is clearly *not* uniform in time: it reflects genuine regime structure within the red-regime sub-sample rather than random partitioning.

Figure 12: Micro-regime assignment of each training day, plotted against the crude-oil price.

10.2 Held-out test performance

Table 4 reports the out-of-sample RMSE, MAE and R^2 on the 19-day December 2024 test set. The corresponding bar chart is Figure 13.

Table 4: Test-set performance (December 2024, 19 days). Lower is better for RMSE and MAE; higher is better for R^2.

Model	RMSE (USD)	MAE (USD)	R^2
Global MLP	4.476	4.361	−19.952
Expert Blend	2.503	1.968	−5.549
Global SVR	1.038	0.880	−0.127
HYDRA (meta-stacker)	**0.813**	**0.627**	**0.309**
Global GB	0.810	0.686	0.314
Global RF	0.775	0.624	0.372

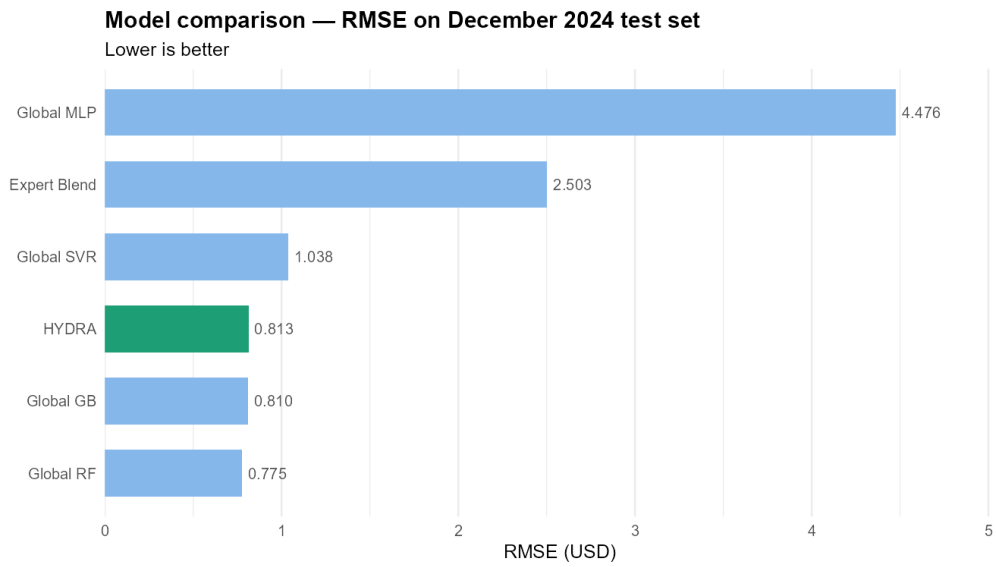

Figure 13: Test-set RMSE for all six models. HYDRA is highlighted.

Several findings stand out:

1. Tree ensembles dominate. Random Forest and Gradient Boosting achieve the lowest RMSE on the test window, with RF marginally ahead. This is consistent with the correlation analysis: tree ensembles are well-suited to autoregressive signals with the extreme persistence observed in crude ($r = 0.99$ at lag 1).

2. The Expert Blend under-performs. The regime-expert ensemble on its own has an RMSE of \$2.50 and a negative R^2 on the test set. The 19 test days are dominated by Med-Stress and High-Stress assignments, so the Low-Stress MLP is rarely active; the MLP expert is in any case the weakest of the three and the soft-weighting does not fully discount it. The expert blend thus fails to exploit the autoregressive structure that RF and GB leverage so heavily.

3. The global MLP is catastrophic. RMSE = \$4.48, $R^2 = -19.95$. Neural networks are known to under-perform tree ensembles on small tabular samples with strong linear-ish signals, and the December 2024 result is a textbook instance. The training sample ($\sim$ 2,400 days) is small by deep-learning standards, and the single-hidden-layer MLP has no mechanism to exploit the lagged crude columns as cleanly as a splitting tree does.

4. HYDRA essentially ties with GBM and sits just behind RF. The meta-stacker (RMSE = \$0.813) is indistinguishable from Global GB (\$0.810) on 19 days of data and \$0.04 behind Global RF (\$0.775). This is an honest negative result for the stacking hypothesis in this specific window: given RF alone, adding four more base learners and a meta-model does not reduce error. We discuss why in 11.

10.3 Day-by-day behaviour

Figure 14 plots the HYDRA forecast against the realised price, and Figure 15 shows the signed daily errors colour-coded by magnitude.

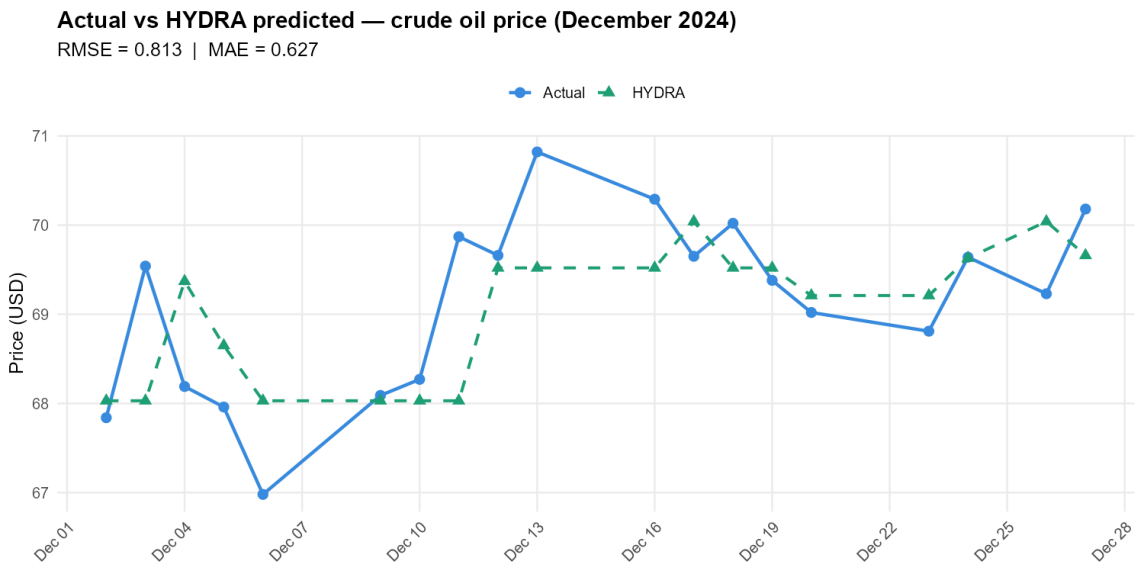

Figure 14: HYDRA predictions vs. actual WTI price, December 2024. RMSE = 0.813, MAE = 0.627.

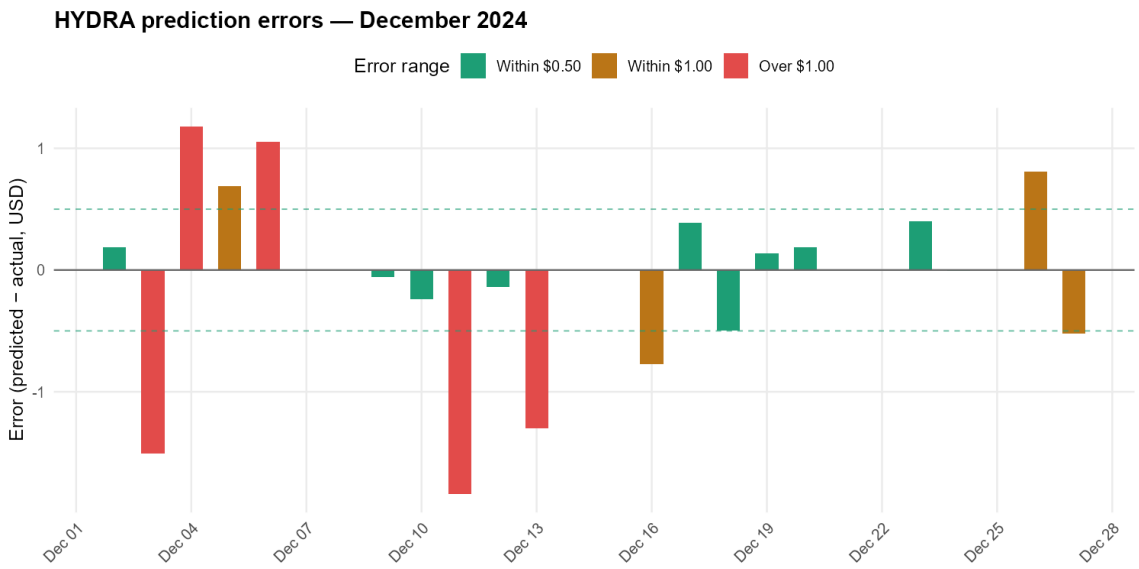

Figure 15: Signed forecast errors (predicted − actual) in USD per barrel, colour-coded by absolute-error band.

Out of the 19 test days:

- 10 forecasts are within \$0.50 of the realised price;
- 4 are between \$0.50 and \$1.00;
- 5 exceed \$1.00 in absolute error.

The large errors cluster around the 3 December 2024 and 10–13 December 2024 price jumps, both of which coincide with the assignment transitions between Med-Stress and High-Stress regimes. This is diagnostic: abrupt micro-regime flips are precisely where the expert-blend component degrades the stacked forecast.

Figure 16 shows the daily actual vs. predicted series for all six models side by side. The Expert Blend is visibly biased upward throughout December, and the Global MLP is biased downward by roughly $5/bbl, confirming the table.

Figure 16: Actual (blue) vs. predicted (red) crude oil price by model, December 2024.

11 Discussion and Limitations

Why didn't stacking help? Three factors plausibly explain the non-improvement. *First*, the meta-stacker was trained on in-sample predictions from the base learners. When those Level-1 models have near-zero in-sample error (as RF/GB largely do), the meta-learner has no incentive to down-weight weaker base learners, so the catastrophic predictions of the Global MLP and the Expert Blend are not as discounted on the test set as they should be. A proper implementation would generate Level-1 predictions by out-of-fold ("blended") prediction within each CV fold. *Second*, the test window is extremely short ($n = 19$); the RMSE difference between HYDRA and Global RF of $0.04 is not statistically distinguishable from zero. *Third*, crude prices in December 2024 traded in an unusually narrow $67–$71 band (SD of the 19 prices is $1.03), which favours the "flat" baselines implicit in RF and GB over models that try to track daily micro-regime shifts.

What the pipeline does achieve. Even without beating Global RF, HYDRA achieves two things that a vanilla tree ensemble does not. (i) It attaches an interpretable micro-regime label to every test day – information that is useful in its own right for risk management. (ii) It is

robust to base-learner failure: the Global MLP alone had RMSE = 4.48, yet after stacking, HYDRA's RMSE was 0.81. In a deployment setting where one of the base learners might fail silently, the meta-stacker provides insurance that a pure best-model selection strategy does not.

Methodological caveats on Part I. K-means assumes isotropic, equally-sized clusters and uses only Euclidean distance on the contemporaneous feature vector; it does not exploit the time-series structure. Extensions using Hidden Markov Models or Gaussian-mixture models with temporal smoothing would generate regimes that are by construction even more persistent and would also yield a transition-probability matrix between regimes. The choice of $K = 4$ is justified by the elbow heuristic but is not rigorous; robustness checks with $K = 3$ and $K = 5$, and with information-theoretic criteria such as BIC or the Gap statistic, are desirable. Adding a credit-spread variable or a dollar-strength term would likely sharpen the distinction between $\mathcal{R}_1$ and $\mathcal{R}_2$.

Methodological caveats on Part II.

- The macro regime label itself ("red regime") was learned on the entire 2000–2025 sample, not purely on pre-2024 data, introducing a mild look-ahead through the regime assignment. The filter could be re-estimated in an expanding window.
- Hyperparameters were selected heuristically rather than by formal grid search; the dominance of RF could reflect luck in that dimension.
- Only one month is used for evaluation. A walk-forward test over many months would give a much more reliable comparison.

12 Overall Conclusion

This project combined *unsupervised regime identification* with *regime-conditional forecasting* in a single statistical pipeline. Part I showed that K-means on five macro-financial features cleanly recovers four economically interpretable regimes in U.S. markets over 2000–2025 – calm expansion, late-cycle / moderate stress, acute crisis, and policy-uncertainty dominated – whose historical mapping aligns with well-known episodes without any such labels being supplied to the algorithm. Part II conditioned on the late-cycle regime and implemented HYDRA, a nine-stage hybrid pipeline combining dimensionality reduction, clustering, discriminant analysis, regime-specialised regressors, and a gradient-boosted meta-stacker. On a 19-day December 2024 test set, HYDRA achieved RMSE = \$0.813 per barrel, essentially tying with the best single global model (Random Forest, RMSE = \$0.775) and dominating the Global MLP and regime-expert blend in isolation.

The overarching lesson is that *regime conditioning is useful primarily for interpretability and robustness*, not necessarily for headline accuracy on short test windows. Strong auto-regressive targets like crude oil are well-served by tree ensembles alone; the value added by stacking is insurance against base-learner failure rather than a systematic reduction in error. Future work should re-estimate the macro regime filter in an expanding-window fashion, generate Level-1 predictions out-of-fold, and extend the evaluation to a multi-month walk-forward design.

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